

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PHD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Lecture 4

Multivariate Dynamics: VAR, Cointegration and VEC

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Vector Autoregression · Granger Causality · Impulse Response Functions · Variance
Decomposition · Structural VAR · Johansen Tests · Vector Error Correction Models

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1. Vector Autoregression

1.1. Definition and estimation

Definition 1.1. The $VAR(p)$ model for an n -dimensional vector $\mathbf{X}_t$ is:

$$\mathbf{X}_t = \boldsymbol{\mu} + \mathbf{A}_1 \mathbf{X}_{t-1} + \cdots + \mathbf{A}_p \mathbf{X}_{t-p} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim WN(\mathbf{0}, \boldsymbol{\Sigma}).$$

$\mathbf{A}_i$ are $n \times n$ coefficient matrices; $\boldsymbol{\Sigma}$ is positive definite.

Each equation is a reduced-form regression of $X_{i,t}$ on lags of all variables. Since all equations share the same regressors, OLS estimated equation-by-equation is efficient (equivalent to GLS). No a priori restrictions on who causes whom are imposed — this was Sims's (1980) critique of large structural models with many debatable exclusion restrictions.

Stationarity condition. All eigenvalues of the companion matrix must lie inside the unit circle. Equivalently, $\det(\mathbf{I} - \mathbf{A}_1 z - \cdots - \mathbf{A}_p z^p) = 0$ must have all roots outside the unit circle.

1.2. Lag length selection

Information criteria for lag selection in VAR:

$$\begin{aligned} \text{AIC} &= \ln |\hat{\boldsymbol{\Sigma}}| + \frac{2pn^2}{T}, \\ \text{BIC} &= \ln |\hat{\boldsymbol{\Sigma}}| + \frac{pn^2 \ln T}{T}. \end{aligned}$$

BIC is preferred for its consistency. Always verify that residuals are white noise after selecting p (multivariate Portmanteau or LM test).

2. Granger Causality

Definition 2.1. In a bivariate $VAR(p)$ for (Y_t, Z_t) : Z_t does **not Granger-cause** Y_t if all lags of Z are jointly zero in the Y -equation:

$$H_0 : A_{12,1} = A_{12,2} = \cdots = A_{12,p} = 0.$$

Test: F -test or χ^2 Wald test with p restrictions.

Granger causality is a *predictive* concept: does past Z improve forecasts of Y beyond what past Y alone provides? It does not imply structural (economic) causation. The test is valid for stationary VAR; for $I(1)$ variables use the Toda–Yamamoto procedure.

3. Impulse Response Functions and Variance Decomposition

3.1. MA(∞) representation and IRF

The stationary VAR(p) has a Wold MA(∞) representation:

$$\mathbf{X}_t = \boldsymbol{\mu} + \sum_{h=0}^{\infty} \boldsymbol{\Psi}_h \boldsymbol{\varepsilon}_{t-h}.$$

The **impulse response function** (IRF) measures the response of $X_{j,t+h}$ to a unit shock in $X_{i,t}$:

$$\text{IRF}_{ij}(h) = [\boldsymbol{\Psi}_h]_{ji}.$$

Identification problem. The reduced-form errors $\boldsymbol{\varepsilon}_t$ are generally correlated ($\boldsymbol{\Sigma} \neq \mathbf{I}$), so “a unit shock to ε_i ” is not well-defined. Two standard approaches:

- (i) **Cholesky decomposition:** $\boldsymbol{\Sigma} = \mathbf{P}\mathbf{P}'$ (lower triangular $\mathbf{P}$). Orthogonalise shocks $\mathbf{u}_t = \mathbf{P}^{-1}\boldsymbol{\varepsilon}_t$. The causal ordering matters; variable listed first is causally prior.
- (ii) **Sign restrictions** (Uhlig, 2005): restrict signs of IRFs based on economic theory without imposing exact zeros. Yields an identified set rather than a point identification.

3.2. Forecast error variance decomposition (FEVD)

FEVD measures the fraction of h -step-ahead forecast error variance of X_j attributable to shocks in X_i . It sums to 100% across all shocks for each variable and horizon.

IRFs are estimated with substantial sampling uncertainty. Always plot bootstrap confidence bands (68% or 90%). Bands that include zero indicate that the response is not statistically distinguishable from zero.

4. The Johansen Cointegration Framework

4.1. From VAR to VEC

A VAR(p) in levels reparameterises as:

$$\Delta \mathbf{X}_t = \boldsymbol{\mu} + \boldsymbol{\Pi} \mathbf{X}_{t-1} + \sum_{i=1}^{p-1} \boldsymbol{\Gamma}_i \Delta \mathbf{X}_{t-i} + \boldsymbol{\varepsilon}_t,$$

where $\boldsymbol{\Pi} = \sum_{i=1}^p \mathbf{A}_i - \mathbf{I}$ and $\boldsymbol{\Gamma}_i = -\sum_{j=i+1}^p \mathbf{A}_j$.

The rank of $\boldsymbol{\Pi}$ determines the cointegration structure:

- $\text{rank}(\Pi) = 0$: no cointegration $\Rightarrow$ VAR in differences.
- $\text{rank}(\Pi) = n$: all variables are $I(0)$ $\Rightarrow$ VAR in levels.
- $0 < \text{rank}(\Pi) = r < n$: r cointegrating vectors $\Rightarrow$ VEC model.

When $r = \text{rank}(\Pi)$: $\Pi = \alpha\beta'$ where β ($n \times r$) contains the cointegrating vectors and α ($n \times r$) contains the adjustment (loading) coefficients.

4.2. Johansen trace and maximum eigenvalue tests

Definition 4.1. Trace test: H_0 : at most r cointegrating vectors vs. more than r :

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^n \ln(1 - \hat{\lambda}_i).$$

Maximum eigenvalue test: H_0 : exactly r vs. $r + 1$:

$$\lambda_{\text{max}}(r, r + 1) = -T \ln(1 - \hat{\lambda}_{r+1}).$$

Proceed sequentially from $r = 0$ upward; stop when H_0 is not rejected.

The five Johansen cases (no constant, restricted constant, unrestricted constant, restricted trend, unrestricted trend) have different critical values and economic interpretations. The most common for $I(1)$ variables with non-zero mean is Case 2 (restricted constant). Misspecifying deterministics changes critical values and may alter rank conclusions.

4.3. Identification of cointegrating vectors

When $r > 1$, individual cointegrating vectors are not identified without further restrictions. Common approaches include normalisation (set $\beta_{11} = 1$), economic theory restrictions (unit elasticities, homogeneity), and exclusion restrictions ($\beta_{ij} = 0$).

5. Applied Examples

5.1. Macro-finance: monetary policy VAR

A trivariate VAR with the federal funds rate, output gap, and inflation is the workhorse model for monetary policy analysis. Cholesky-identified IRFs (policy rate ordered last) show the response of output and inflation to a contractionary monetary shock: output falls after 4–6 quarters; inflation falls more slowly (price puzzle in some specifications).

5.2. Pairs trading

For two stock prices (P_{1t}, P_{2t}) both $I(1)$: test $r = 1$ cointegrating vector $[1, -\theta]$. The spread $P_{1t} - \theta P_{2t}$ is stationary. The VEC model governs how each price adjusts to restore equilibrium after a deviation — the basis for statistical arbitrage.

6. Core References

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